Generalization in Diffusion Models arises from Geometry-Adaptive Harmonic Representations

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Motivation

Key Question

- Do diffusion models really **learn the data distribution**, or do they simply **memorize training samples**?
- Why do they generalize surprisingly well despite the **curse of dimensionality**?

Observation

- High-dimensional density estimation is usually hard
- Yet diffusion models generate realistic, diverse samples

Main Contributions

This paper provides a theoretical explanation for why diffusion models generalize.

1. **Denoising = Shrinkage in Adaptive Basis**
   - DNN denoisers perform shrinkage along learned basis directions
   - Links optimal denoising to sparsity and eigenstructure
2. **Geometry-Adaptive Harmonic Bases (GAHB)**
   - DNNs learn basis functions aligned with image geometry
   - Explains near-optimal performance on natural images
3. **Inductive Bias Explains Generalization**
   - Generalization comes from strong inductive bias toward GAHBs
   - Mismatch with data (e.g. low-dimensional manifolds) leads to suboptimality

Diffusion as Denoising

Noisy Observation

$$y = x + z, \quad z \sim \mathcal{N}(0, \sigma^2 I)$$

- x: clean data y: noisy observation

Denoising

$$f_\theta(y) \approx x$$

Score

$$\nabla \log p_\sigma(y)$$

Miyasawa / Tweedie

$$\nabla \log p_\sigma(y) = \frac{\mathbb{E}[x|y] - y}{\sigma^2}$$

$$f^*(y) = \mathbb{E}[x|y]$$

- $f^*(y)$: optimal denoiser (MSE)

Denoising Error and Generative Modeling

KL Bound

$$D_{KL}(p(x) \,\|\, p_\theta(x)) \leq \int_0^\infty \mathbb{E}_y \left[\|\nabla \log p_\sigma(y) - s_\theta(y)\|^2 \right] \sigma \, d\sigma$$

Interpretation

- score error $=$ denoising error
- small denoising error $\Rightarrow$ small KL

$$\text{good denoiser} \implies \text{good generative model}$$

Numerical Experiment

Setup

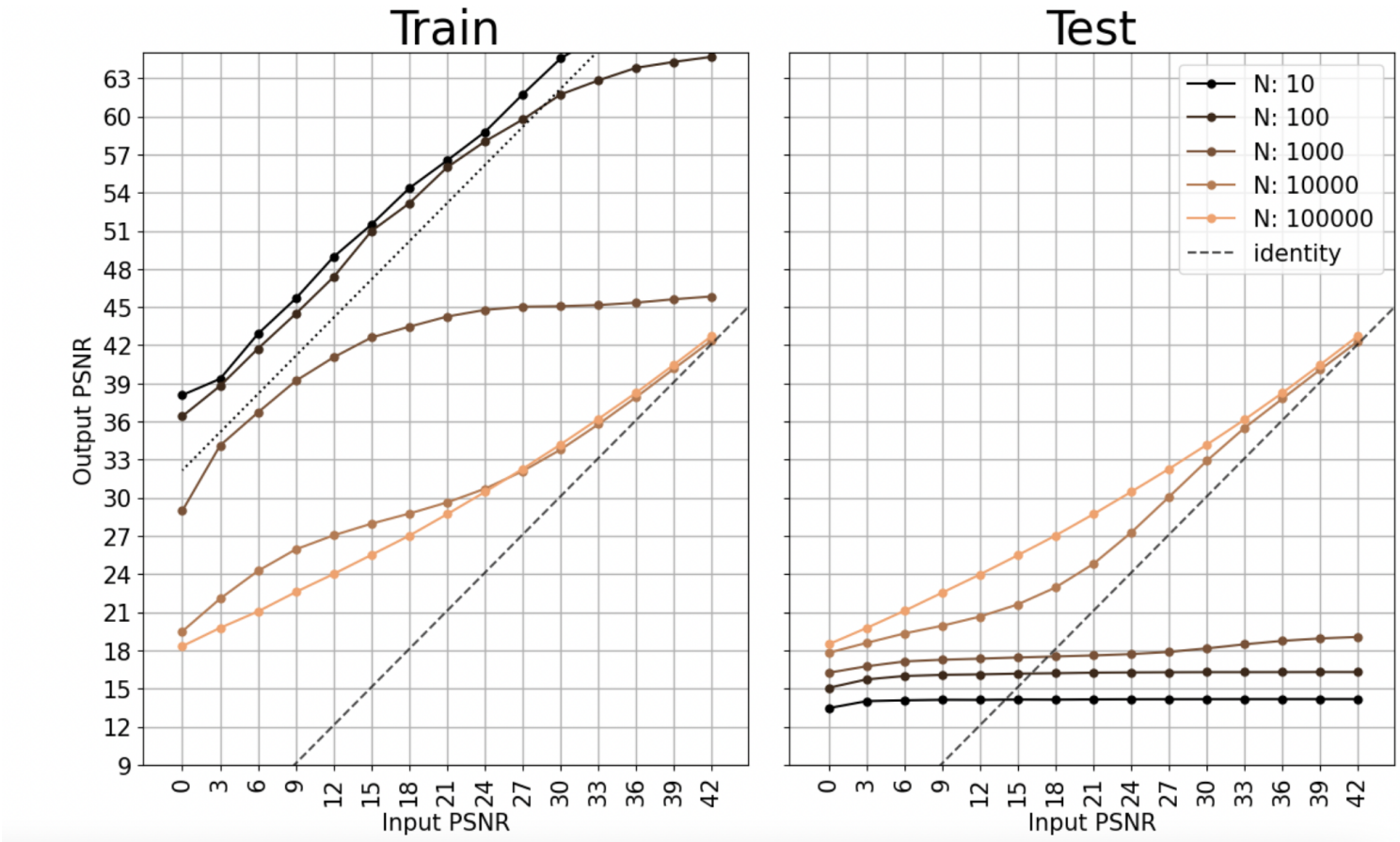
- grayscale CelebA faces
- two disjoint training sets (A / B)
- UNet denoisers trained with

$$N \in \{10,\ 100,\ 1000,\ 10000,\ 100000\}$$

Evaluation

- one-shot denoising (PSNR)
- generation from same noise seed

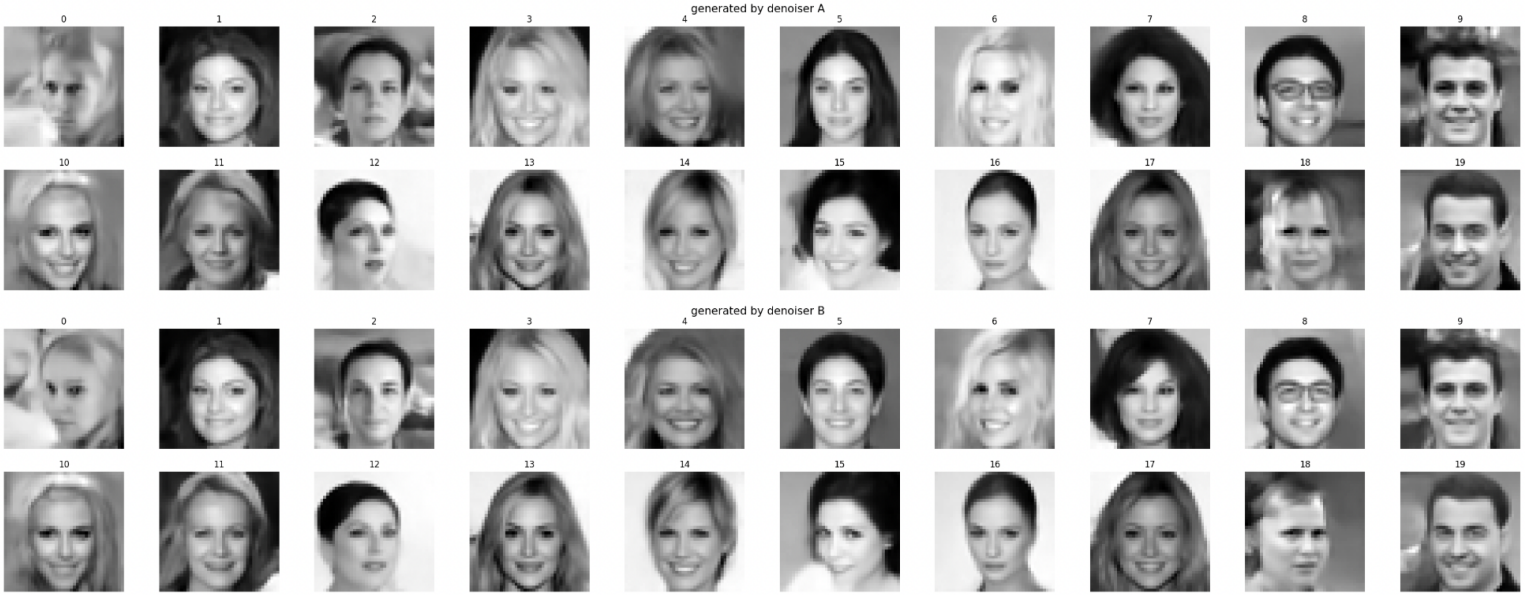
PSNR Curves



Conclusion

- larger $N \Rightarrow$ test PSNR $\uparrow$
- train/test gap $\downarrow$
- transition from memorization to generalization

Generated Samples



Conclusion

- small N: samples close to training images
- large N: samples become independent of specific data
- models trained on A / B produce similar outputs

Denoising as Shrinkage

Local Linearization of Denoiser

$$f(y) \approx \nabla f(y) y$$

Eigen-decomposition

$$\nabla f(y) = \sum_k \lambda_k(y) e_k(y) e_k(y)^\top$$

Shrinkage Form

$$f(y) = \sum_k \lambda_k(y) \langle y, e_k(y) \rangle e_k(y)$$

- $e_k(y)$: adaptive basis
- $\lambda_k(y)$: shrinkage factors

Optimal Denoiser

Optimal Solution (MSE)

$$f^*(y) = \mathbb{E}[x|y]$$

Jacobian

$$\nabla f^*(y) = \sigma^{-2}\mathrm{Cov}[x|y]$$

Eigen-decomposition

$$\mathrm{Cov}[x|y] = \sum_k \lambda_k \, e_k e_k^\top$$

- eigenvectors: principal directions of variation of $x|y$
- eigenvalues: uncertainty (variance) along each direction

Oracle Denoising Error

Fixed basis + oracle shrinkage

$$\lambda_k = \lambda_k(x)$$

Denoising error (MSE)

$$\mathbb{E}_x \left[\sum_k \left((1 - \lambda_k(x))^2 \langle x, e_k \rangle^2 + \lambda_k(x)^2 \sigma^2 \right) \right]$$

Interpretation

- first term: signal loss (bias)
- second term: noise contribution (variance)

Optimal shrinkage

$$\sum_k \min \left(\langle x, e_k \rangle^2, \sigma^2 \right)$$

Sparsity and Error Rate

Sparsity assumption

$$\langle x, e_k \rangle^2 \sim k^{-(\alpha+1)}$$

Resulting rate

$$\text{MSE} \sim \sigma^{\frac{2\alpha}{\alpha+1}}$$

$$\text{PSNR slope} = \frac{\alpha}{\alpha+1}$$

- faster decay $\Rightarrow$ sparser representation
- sparsity $\Rightarrow$ smaller denoising error

Adaptive Bases and Near-Optimality

From Fixed to Adaptive Basis

- allow basis to depend on data: $e_k = e_k(y)$
- aim: obtain a **sparser representation** of x

Oracle Adaptive Basis

$$\langle x, e_k(x) \rangle^2 \sim k^{-(\alpha+1)}$$

$$\Rightarrow \quad \text{PSNR slope} = \frac{\alpha}{\alpha + 1}$$

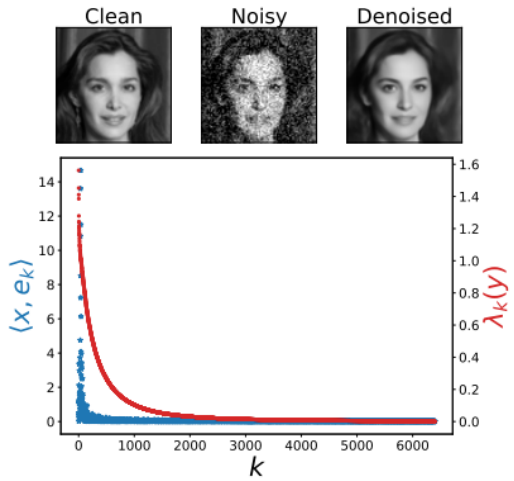
Practical Constraint

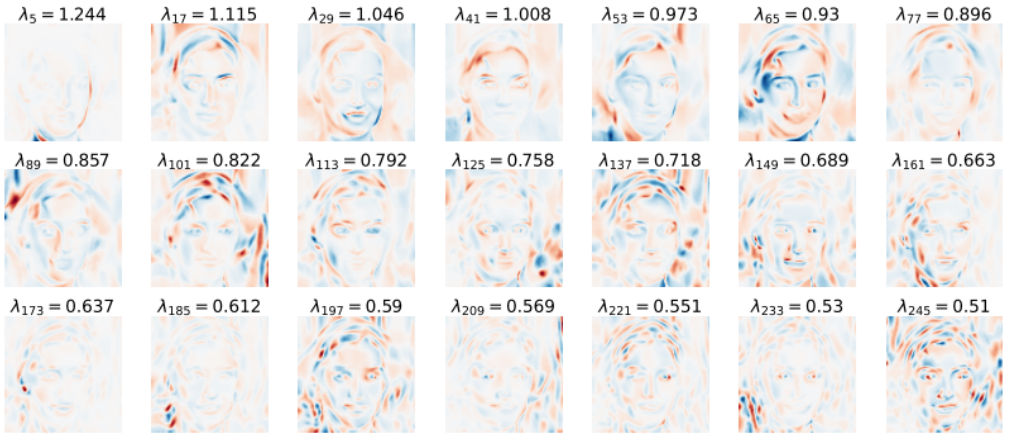
- cannot use $e_k(x)$ (depends on unknown x)
- restrict to a dictionary of bases

Key Result

- adaptive basis $e_k(y)$ can achieve the same rate
- near-optimal despite noise

Geometry-Adaptive Harmonic Bases (GAHB)





Observation

- $e_k(y)$: adaptive and oscillatory
- aligned with contours and smooth regions

Conclusion

DNN learns a geometry-adaptive harmonic basis

Why GAHB Matters

Hypothesis of the paper

- DNN denoisers are inductively biased toward GAHBs

If true, then:

- when GAHBs are optimal for the data, DNNs should be near-optimal
- when GAHBs are not optimal, DNNs should become suboptimal

Testing strategy

- C^α images: favorable case
- low-dimensional manifolds: unfavorable case
- shuffled faces: locality destroyed

Favorable Case: C^α Images and Bandlets

C^α images

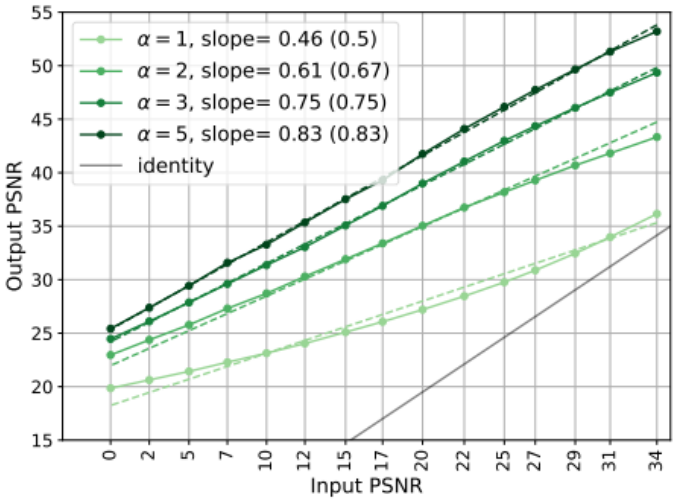
- regular contours on regular backgrounds
- controlled by regularity parameter α

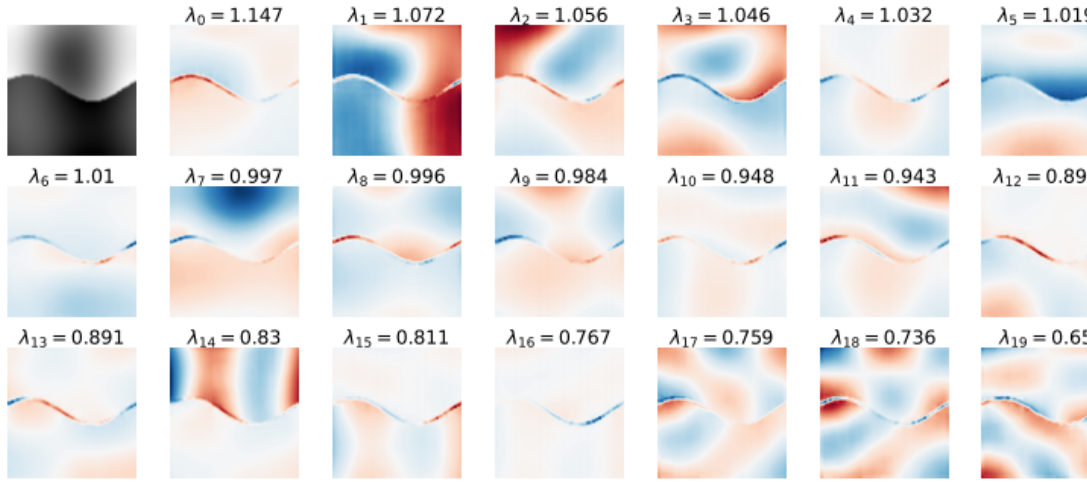
Known theory

- optimal sparse representations are given by **bandlet bases**
- bandlets are harmonic functions adapted to image geometry

$$\langle x, e_k \rangle^2 \sim k^{-(\alpha+1)} \qquad \Longrightarrow \qquad \text{PSNR slope} = \frac{\alpha}{\alpha+1}$$

Near-Optimality on C^α Images





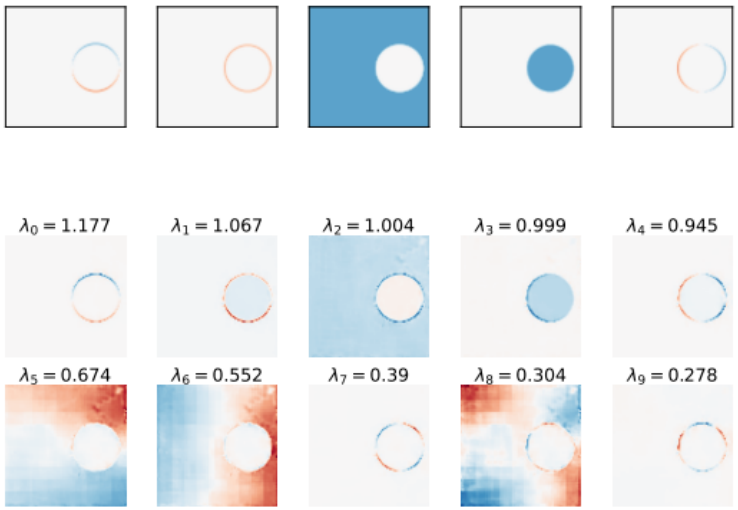
Observation

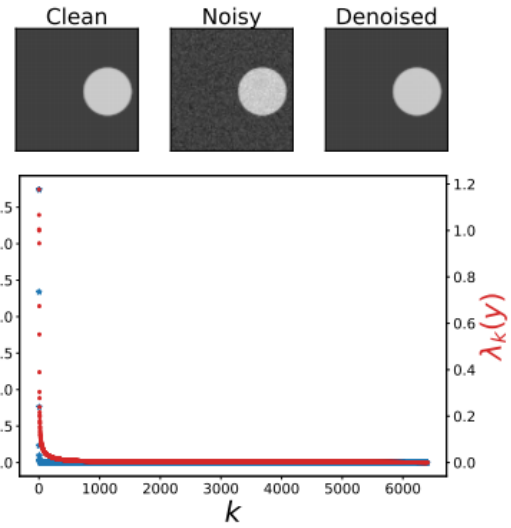
- PSNR slope $\approx \frac{\alpha}{\alpha+1}$ (optimal)
- learned eigenvectors resemble bandlets / GAHBs

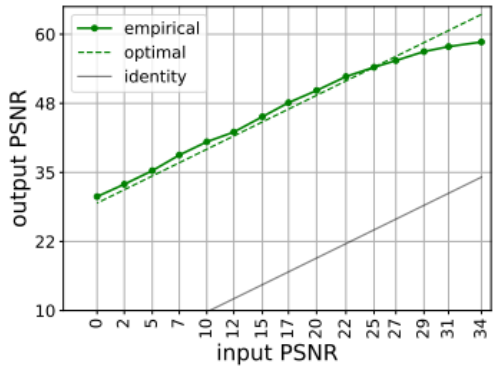
Conclusion

DNN achieves near-optimal denoising on C^α images

Failure on Low-Dimensional Manifolds







Observation

- optimal: few directions $\Rightarrow$ slope $= 1$
- DNN: extra directions $\Rightarrow$ slope < 1

Conclusion

bias mismatch $\Rightarrow$ suboptimal denoising

Additional Evidence: Shuffled Faces

Shuffled faces

- pixels are permuted with a common random permutation
- local image geometry is destroyed

Implication

- the optimal basis is no longer harmonic / local
- DNN inductive bias is no longer aligned with the data

$$\text{destroy locality} \implies \text{worse denoising performance}$$

Main Takeaways

- diffusion can be analyzed through **denoising**
- denoising can be viewed as **shrinkage in an adaptive basis**
- DNNs implicitly learn **geometry-adaptive harmonic bases**
- when this bias matches the data, performance is near-optimal
- when it does not match, performance becomes suboptimal

$$\text{Generalization} \approx \text{Inductive Bias} \approx \text{GAHB}$$

Appendix: Proof of the Miyasawa Relation

Goal

We want to derive the relationship between the score and the MSE denoiser:

$$\mathbb{E}[x|y] = y + \sigma^2 \nabla \log p(y)$$

Setup

$$y = x + z, \qquad z \sim \mathcal{N}(0, \sigma^2 I)$$

so that

$$p(y|x) = \mathcal{N}(x, \sigma^2 I)$$

- x: clean signal
- y: noisy observation
- $p(y)$: noisy-data density

Step 1: Marginal Density of the Noisy Observation

By marginalization,

$$p(y) = \int p(x)\, p(y|x)\, dx$$

This means that the noisy density is obtained by averaging the Gaussian likelihood over the clean-data distribution.

Interpretation

- $p(y)$ is the density of noisy samples
- it is the clean distribution blurred by Gaussian noise

Step 2: Differentiate the Log-Density

We start from

$$\nabla \log p(y) = \frac{\nabla p(y)}{p(y)}$$

Using

$$p(y) = \int p(x)\, p(y|x)\, dx$$

we differentiate with respect to y:

$$\nabla p(y) = \int p(x)\, \nabla_y p(y|x)\, dx$$

Hence,

$$\nabla \log p(y) = \frac{\int p(x)\, \nabla_y p(y|x)\, dx}{p(y)}$$

Step 3: Rewrite Using Bayes' Rule

Write

$$\nabla_y p(y|x) = p(y|x)\, \nabla_y \log p(y|x)$$

Then

$$\nabla \log p(y) = \frac{\int p(x)\, p(y|x)\, \nabla_y \log p(y|x)\, dx}{p(y)}$$

Using Bayes' rule,

$$p(x|y) = \frac{p(x)p(y|x)}{p(y)}$$

we obtain

$$\nabla \log p(y) = \int p(x|y)\, \nabla_y \log p(y|x)\, dx$$

Therefore,

$$\boxed{\nabla \log p(y) = \mathbb{E}[\nabla_y \log p(y|x) \mid y]}$$

Step 4: Use the Gaussian Likelihood

Since

$$y|x \sim \mathcal{N}(x, \sigma^2 I)$$

we have

$$\log p(y|x) = -\frac{1}{2\sigma^2}\|y - x\|^2 + C$$

Differentiating with respect to y gives

$$\nabla_y \log p(y|x) = -\frac{1}{\sigma^2}(y - x)$$

Substituting into the previous identity:

$$\nabla \log p(y) = \mathbb{E}\left[-\frac{1}{\sigma^2}(y - x) \mid y\right]$$

Step 5: First-Order Miyasawa Relation

We simplify:

$$\nabla \log p(y) = -\frac{1}{\sigma^2}(y - \mathbb{E}[x|y])$$

or equivalently,

$$\nabla \log p(y) = \frac{1}{\sigma^2}(\mathbb{E}[x|y] - y)$$

Rearranging gives the **first-order Miyasawa relation**:

$$\boxed{\mathbb{E}[x|y] = y + \sigma^2 \nabla \log p(y)}$$

Interpretation

- posterior mean = observation + score correction
- MSE denoising is equivalent to score estimation

Step 6: Differentiate Again

Now we derive the second-order relation.
Starting from

$$\nabla \log p(y) = \mathbb{E}[\nabla_y \log p(y|x) \mid y]$$

differentiate again with respect to y:

$$\nabla^2 \log p(y) = \mathrm{Cov}[\nabla_y \log p(y|x) \mid y] + \mathbb{E}\left[\nabla_y^2 \log p(y|x) \mid y\right]$$

This is the second-order analogue of the previous identity.

Step 7: Plug the Gaussian Derivatives

For the Gaussian likelihood,

$$\nabla_y \log p(y|x) = -\frac{1}{\sigma^2}(y - x), \quad \nabla_y^2 \log p(y|x) = -\frac{1}{\sigma^2} I$$

Therefore,

$$\mathrm{Cov}[\nabla_y \log p(y|x) \mid y] = \frac{1}{\sigma^4}\mathrm{Cov}[x|y]$$

and

$$\mathbb{E}\left[\nabla_y^2 \log p(y|x) \mid y\right] = -\frac{1}{\sigma^2} I$$

Substituting into the previous formula:

$$\nabla^2 \log p(y) = \frac{1}{\sigma^4}\mathrm{Cov}[x|y] - \frac{1}{\sigma^2} I$$

Step 8: Second-Order Miyasawa Relation

Rearranging,

$$\boxed{\mathrm{Cov}[x|y] = \sigma^2\Big(I + \sigma^2\nabla^2 \log p(y)\Big)}$$

This is the **second-order Miyasawa relation**.

Interpretation

- first order: score gives the posterior mean
- second order: Hessian gives the posterior covariance

So both the denoiser and its local uncertainty structure are encoded in the noisy density.

Step 9: MSE Error Identity

The optimal denoiser is

$$f^*(y) = \mathbb{E}[x|y]$$

Its MSE is

$$\mathbb{E}\|x - \mathbb{E}[x|y]\|^2$$

Using the identity

$$\|v\|^2 = \mathrm{tr}(vv^\top)$$

we get

$$\mathbb{E}\|x - \mathbb{E}[x|y]\|^2 = \mathbb{E}[\mathrm{tr}\,\mathrm{Cov}[x|y]]$$

Thus, the optimal denoising error is exactly the expected posterior variance.

Appendix Summary

First-order relation

$$\boxed{\mathbb{E}[x|y] = y + \sigma^2 \nabla \log p(y)}$$

Second-order relation

$$\boxed{\mathrm{Cov}[x|y] = \sigma^2 \left(I + \sigma^2 \nabla^2 \log p(y) \right)}$$

Consequence

- score $\Longleftrightarrow$ optimal denoiser
- Hessian $\Longleftrightarrow$ posterior uncertainty
- diffusion can be studied through denoising